Cheaper LLMs as AMR literature screeners

Executive Summary

Hyam Ali · Dr Fabrizio Costa · Dr Shiqiang Wang | University of Exeter · Google DeepMind Research Ready | September 2026

Research question: Can a cheaper language model stand in for Claude Sonnet 5 when screening PubMed abstracts, without changing which WHO GLASS pathogen–antibiotic pairs come out as research priorities?

89	73,857	711,925	12	2
pathogen–antibiotic pairs	paper–pair screens	LLM decisions	models tried	acceptance rules

1. Key Findings

1. **The acceptance rule matters more than which model you use.** Under strict-only, any two candidates put 22–52% of pairs in different quadrants, compared with just 11–16% under strict-or-yes. Relaxing the rule helps every model; GPT-3.5, for example, goes from ρ 0.62 to 0.92.
2. **With the permissive rule, most of the agreement comes from publication volume.** If we skip the LLM entirely and accept every abstract, we still get $\rho = 0.95$. Gemini 3.5 Flash-Lite (0.97) is the only model that does better, and both GPT models do worse.
3. **With the strict rule, the choice of model decides the outcome, and even the best one misses emerging pairs.** Gemini 3.5 Flash-Lite reaches ρ 0.87 against 0.67 without an LLM, yet it still moves 38% of pairs into a different quadrant and finds only 2 of the 13 emerging pairs.
4. **Agreeing on individual papers is not the same as agreeing on the ranking.** Gemini is the only model with moderate per-paper agreement (κ 0.58). Gemma 3 270M is no better than chance ($\kappa = -0.01$), and yet it ranks the pairs at $\rho = 0.83$.

2. Method

Every model screened all 73,857 paper–pair items as strict, yes or no. The accepted counts for each pair and year go into a pooled ridge regression, and the residual from that regression is the Scientific Interest (SI) signal. A damped 5-year forecast then puts each pair into a priority quadrant. We compared each model with Sonnet 5 at the paper level (Cohen's κ) and at the pair level (Spearman ρ and quadrant changes), under the **strict-only** and **strict-or-yes** rules and against a **no-LLM baseline** that accepts every abstract.

3. Cost versus Fidelity

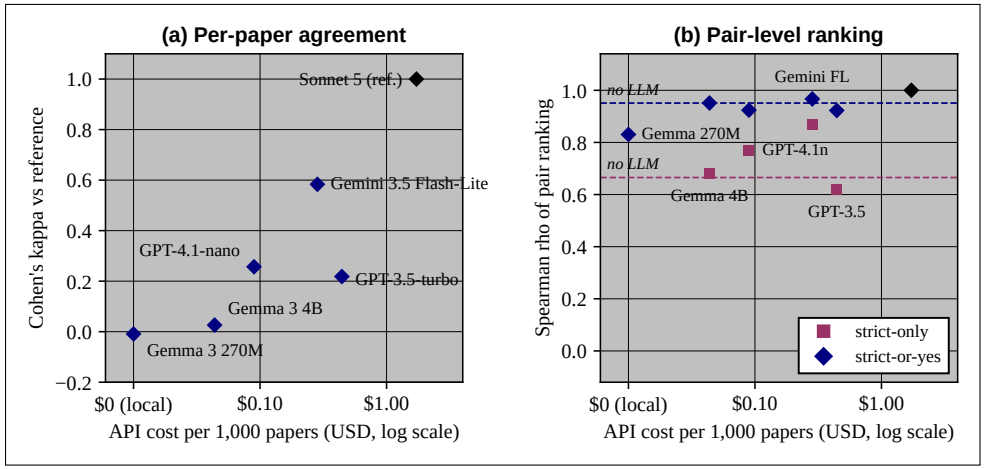


Figure 1. Agreement with the reference against API cost. Dashed lines mark the no-LLM baseline for each rule. Gemma 3 270M has no strict-only value because its SI signal is constant.

Table 1. Screening cost by model.

Model	$ per 1,000 papers	Full corpus	vs. ref.	Kappa
Claude Sonnet 5 (ref.)	$1.72	$127.22	100%	1.00
Gemini 3.5 Flash-Lite	$0.28	$20.97	16%	0.58
GPT-4.1-nano	$0.09	$6.62	5%	0.26
GPT-3.5-turbo	$0.44	$32.74	26%	0.22
Gemma 3 4B	$0.04	$3.23	2.5%	0.03
Gemma 3 270M	$0.00	$0.00	0%	-0.01

Prices are list prices from Jul–Aug 2026, with Sonnet 5 on the Batch API (50% off). The full corpus is 73,849 screens. The local model costs $0 in API fees but took ∼84 h on one GPU.

Recommendation. Gemini 3.5 Flash-Lite is the only substitute that beats the no-LLM baseline under both rules, and it costs 16% of the reference ($21 against $127 for the full corpus). The acceptance rule should be reported as a main design choice, and any strict-or-yes result should be compared with the no-LLM baseline.

Before submission: we still have no human ground truth, and 6.6% of the reference screens came back empty (probably because adaptive thinking hit the 60-token cap), so these need re-running. The work also rests on a single prompt, and the size ladder covers only 3 of the 89 pairs.